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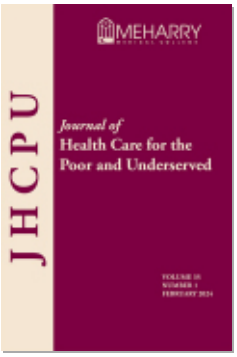

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ORIGINAL PAPER

A Review of Disparities in Outcomes of Hospitalized Patients with Limited English Proficiency: The Importance of Nursing Resources

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Abstract: Language barriers significantly affect communication between patients and health care staff and are associated with receipt of lower-quality care. Registered nurses are well positioned members of the health care team to reduce and eliminate disparities for patients with limited English proficiency (LEP). Current evidence recommends nurses use interpreters or translation devices to overcome language barriers; however, these recommendations fail to recognize that structural system-level factors, such as unsupportive work environments and poor nurse-to-patient staffing ratios, reduce nurses' ability to implement these recommendations. The Quality Health Outcomes Model (QHOM) is a useful framework for understanding relationships between hospital systems, the delivery of care interventions, and patient outcomes. The goal of this manuscript is to use the QHOM and existing empirical evidence to present a new perspective on the long-standing clinical challenge of reducing language-related health outcome disparities by considering the context in which nurses deliver patient care.

Key words: Limited English proficiency, LEP, language barriers, Quality Health Outcomes Model.

Nearly 26 million individuals in the United States (U.S.) have limited English proficiency (LEP), defined as the ability to speak English "less than very well."[1] In the U.S., individuals with LEP have a legal right to access health care in their preferred language. Title VI of the Civil Rights Act of 1964 prohibits discrimination based on race, color, or national origin and mandates the provision of interpreter services by agencies receiving federal financial assistance, which includes hospitals that receive Medicaid and Medicare reimbursement.[2] Language barriers significantly affect com-

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munication between patients and health care staff and are associated with receipt of lower-quality care. Individuals with LEP experience lower satisfaction with care,[3,4] longer lengths of stay,[5,6] and increased hospital readmissions[7,8] compared with those who are English-proficient. These health outcome disparities are often attributed to a lack of proper interpreter use which results in language discordant care, which Batista and colleagues define as occurring when a patient and their provider lack proficiency in a shared language.[9] However, we propose here that a central problem underlying these disparities is in the inadequacies of organizational-level structures that are foundational to nurses in caring for patients.

Role of the Nurse in Caring for Patients with LEP

Although research has demonstrated worse outcomes for LEP patients,[5,7,10–12] little research has focused on the role registered nurses play in reducing and eliminating these disparities. As nurses are often the first and most consistent point of contact for patients and their families during hospitalization, they are well positioned to efficiently assess, monitor, and intervene as needed for patients with LEP. However, communication barriers can significantly impede care delivery.[13,14]

Interpreter services have been identified as a key intervention in overcoming language barriers between patients and health care staff; they can raise the quality of care that LEP patients receive.[15] Language concordant care can also improve patient satisfaction and increase LEP patients' comfort.[16] Previous research recommends nurses' use of interpreters or translation devices to overcome language barriers[17] but fails to recognize the influence system-level factors may have on the implementation of such interventions. Existing health services research demonstrates that unsupportive work environments, limited time, and poor nurse-to-patient staffing ratios can affect nurses' capacity to provide quality care such as appropriate patient-nurse communication and effective patient teaching.[18–22] The effect of these system-level factors may be even stronger when caring for patients with LEP, given their complex communication needs that require additional nursing care time.[23]

Objective

The objective of this commentary is to present a new perspective for understanding what system-level factors, and thus possible solutions, might underlie hospital outcomes disparities for LEP patients. Informed by a review of extant literature, we discuss the theoretical relationships between organizational features of hospital nursing care (e.g., staffing ratios and nurse work environments) and outcomes for hospitalized patients with LEP.

Quality Health Outcomes Model

Overview. The Quality Health Outcomes Model (QHOM) by Mitchell is a useful framework for understanding the role hospital systems have in the delivery of nursing care.[24] The QHOM demonstrates relationships among four elements: system, interven-

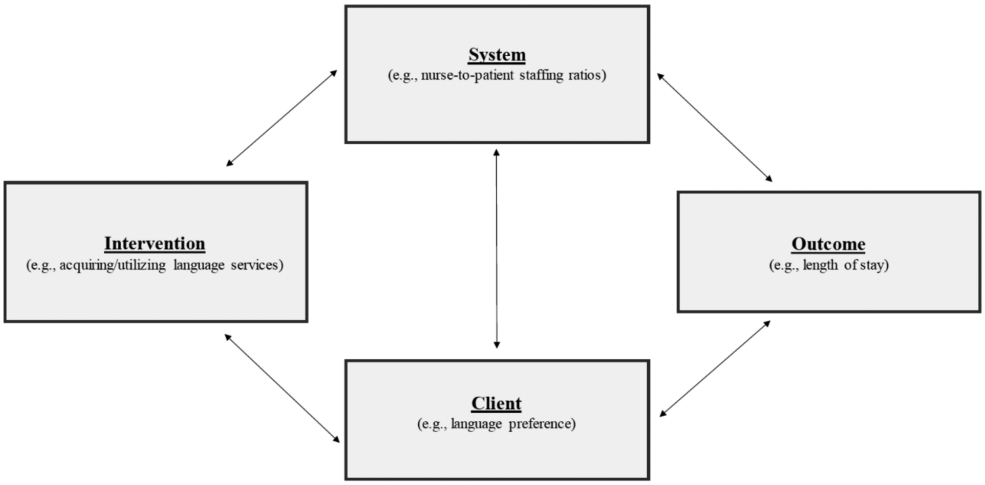

Figure 1. This conceptual model is an adaptation of the Quality Health Outcomes Model,[24] applying its components to the context of language barriers within hospitals.

tion, outcome, and client. *System* refers to the settings where care occurs, including both material and human resources, and considering the organization's attributes such as size and location. *Intervention* refers to the action of giving and receiving care that affects health, such as performing a patient assessment, providing patient education, and administering medications. *Outcomes* can refer to the health status, functional ability, health care costs, or satisfaction of patients. Lastly, *client* refers to an individual, group, or population receiving health care services and their characteristics such as age or co-morbidities.[24] A key feature of this model is that it demonstrates that hospital systems act as moderators of an intervention's effect on outcomes, as demonstrated in Figure 1.

Existing research on language barriers in the hospital setting has largely focused on the intervention (e.g., examining effects of nurse interpreter use) and outcome (e.g., outcome disparities of LEP patients compared with English-proficient patients) components of the QHOM, but has failed to empirically study the systems component and how it influences interventions and outcomes.

Review of the Literature Using the QHOM Framework

Methods. A review of the literature was performed using the Cumulative Index to Nursing and Allied Health Literature (CINAHL) in June 2022. Included articles were required to have been peer-reviewed, to be in English, and to have been disseminated between January 2007 and May 2022 to ensure relevant and up to date literature from the past 15 years. The search was expanded to hospitals within and outside of the United States, given the overall lack of literature on this topic. The titles were first screened for relevance to hospital nursing resources, nursing care delivery, and outcomes for patients with LEP. Then, abstracts were reviewed for articles with relevant titles.

Outcomes. A strong body of research has demonstrated that patients with LEP

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experience worse hospital outcomes, including more readmissions, adverse events, longer lengths of stay, mortality, and decreased patient satisfaction compared to their English-proficient counterparts, especially when interpreter services or bilingual care providers are not used properly.

Higher risk of readmission and adverse events. Individuals with LEP experience a higher risk of readmission,[7,8,24] which may be caused in part by unaddressed language barriers contributing to misdiagnoses and delayed treatment.[25] Language barriers can threaten patient safety, specifically with respect to medication errors that result from communication difficulties.[26,27] A study of six U.S. hospitals examined variation in adverse event occurrence between LEP and non-LEP patients. The study found that 49.1% of LEP patients experienced adverse events (i.e., medication errors, patient falls, skin breakdown, injury during treatment, or equipment/instrument issues) involving some physical harm whereas only 29.5% of English proficient patients experienced adverse events that resulted in physical harm.[28] Many LEP patients experience adverse events due to the use of ad hoc interpreters[29] such as friends, family members, or other unqualified bilingual staff who are improperly used as interpreters. Ad hoc interpreter use has been shown to contribute to mistranslations between patients and providers.[30]

Increased length of stay and mortality rates. Studies have demonstrated that LEP patients tend to have longer lengths of stay (LOS) in comparison with English-proficient patients.[5,6] John-Baptiste et al. found that LEP patients had longer hospital stays for seven of 23 medical conditions including unstable coronary syndromes, chest pain, coronary artery bypass grafting, stroke, and diabetes mellitus, with LOS differences ranging from approximately 0.7 to 4.3 days.[6] Similarly, a study of over 3,000 LEP patients showed that patients who did not receive professional interpretation at admission and/or discharge had an increase in their LOS between 0.75 and 1.47 days, compared with LEP patients who had an interpreter on both day of admission and discharge, demonstrating the importance of using interpreter services.[31] Patient-provider language discordance is also associated with an increased risk of death, as demonstrated by a retrospective cohort study of patients receiving tuberculosis treatment.[32]

Decreased patient satisfaction. Language barriers are associated with worse LEP patient experiences, particularly regarding communication with health care providers.[4] One study found that compared with English-proficient breast cancer survivor Latinas, Spanish language-proficient Latinas were less satisfied with their care and patient-provider communication and had lower reports of emotional and functional wellbeing.[3] Limited English proficiency patients often report feeling frustrated, confused, and worried about receiving care when language barriers are present.[33]

Interventions. Nurses are responsible for the implementation of nursing interventions, which include building a nurse-patient relationship, performing detailed assessments and critical screening, and providing patient education, including information related to medications. All of these require clear and effective communication, which is hindered when a language barrier is present and can lead to poorer outcomes.[34]

Nurse–patient relationship. A strong nurse-patient relationship is essential for effective, patient-centered care delivery. Research shows that a lack of shared colloquial language negatively affects nurses' abilities to understand patients and establish verbal

communication, which nurses describe as a usual and important aspect of their care.[35] Language barriers hinder nurses' ability to develop rapport, a critical step in creating a strong nurse-patient relationship, because they are simply unable to understand their patients.[36] Nurses have reported that the relationship they share with a patient with LEP is less engaging and less personal than their relationships with non-LEP patients and that the effort required to access resources to improve the nurse-patient relationship (e.g., translator services) often leaves nurses feeling exhausted both emotionally and mentally.[37] Nurses express a desire to connect with LEP patients but say they often struggle to do so.[14] Nurses reported language barriers as a challenge to the nurse-patient relationship not present in caring for English-speaking patients. To build the nurse-LEP patient relationship requires additional time, which nurses do not always have to spare given their already heavy workloads.[14]

Missed care. Missed care can be defined as an act of omission, further defined as failing to do the right thing for a patient[38] and is a concern for patient safety because it can contribute to poor clinical outcomes.[39] Existing research demonstrates that one of the most cited reasons for missed nursing care is miscommunication.[38,40] If missed communication occurs for English-proficient patients, it is even more likely that LEP patients, given their complex communication needs, are vulnerable to experiencing miscommunication, subsequent missed care, and poor clinical outcomes. Reppas-Rindlisbacher et al. demonstrated that many nurses did not feel confident of their ability to accurately assess delirium in patients with LEP using the Confusion Assessment Method (CAM) screening tool due to worries regarding the accuracy of the assessment or that administering the assessment might worsen confusion in LEP patients.[41] Nurses experiencing barriers to using screening tools for patients with LEP can lead to missed care and patients being overlooked in the screening of clinical conditions such as delirium. Other nursing care interventions that are likely to be missed due to language barriers include patient teaching, discharge planning, and individual interventions.[40] Gelete et al. demonstrated that language barriers create barriers to providing patients with important health-related discharge information and obtaining consent, both of which are critical aspects of the nursing care delivery process.[42] Similarly, clinicians in Australia identified that language barriers, particularly among refugees, altered the health care delivery process when it came to understanding patients' symptoms and subsequently performing an appropriate assessment.[43]

Delays in provision of care. Language barriers are known to cause delays in the provision of nursing care due to communication difficulties and long wait times to access interpreter services.[13,44,45,46] This can cause further delays in nurses establishing contact and acquaintance with patients.[46] Nurses have also highlighted that an inability to access interpreters may result in cancellation of appointments or unnecessary delays in nursing care provision, which could subsequently lead to an increased length of stay for the patient and increased expenses.[47] A randomized control trial demonstrated the benefit of an admission service that ensured language concordance between nurses and patients, which reduced the overall time taken to admit a patient to the hospital.[48]

System. While several studies highlight system-level barriers such as heavy workloads, poor staffing, unsupportive work environments, and other nursing workplace

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issues, including availability of interpreter services, to the delivery of high-quality care to patients with LEP, no current literature has examined the direct associations between system-level hospital nursing resources and LEP patient outcomes.

Need for time and human resources (staffing). Due to patients with LEP requiring additional nursing time,[23] Coleman and Angosta recommend that LEP status be considered in terms of workload assignments, with nurses caring for patients with LEP being given a lighter assignment.[14] One study demonstrated that clinicians could not access interpreter services when caring for patients with LEP due to their lack of time to wait for interpreter arrival, with one stating "the time is the biggest barrier of anything."[44] In a study of nurses' perceptions of an existing telephone-based remote interpretation technology compared with perceptions of a video remote interpretation (VRI) system, nurses preferred the VRI system because it was more time efficient and aided their overall care delivery process.[49] Furthermore, nurses have reported system-related frustrations in having insufficient time to spend with their patients due to the lack of human resources available.[46] Issues with the provision of adequate staffing levels and staff having no time are commonly recognized in research on language barriers. One qualitative study demonstrated that the increased workload that ensues from a shortage of nursing staff results in limited time for patient care and overlooking cultural or linguistic considerations relevant to the patients' care needs.[42] Another study found that although nurses recognized language barriers as affecting their ability to effectively communicate with patients, they also highlighted that being overworked, experiencing understaffing of nurses, and lacking time were the most important barriers for nurses in establishing effective communication with their patients.[35]

Interpreter services. Using interpreter services is a strongly recommended intervention in overcoming language barriers as interpreter services facilitate language concordance between patients and health care staff.[15,49] A systematic review by Diamond et al. showed that language-concordant care, which occurs when patients and health care providers share a common language, had a positive effect on the quality of care patients received, specifically related to diabetes care, pain management, and cancer care.[50] However, even when language services are offered, nurses experience system-level barriers to accessing them. Although many nurses recognize the value of using interpreter services, they have stated their experiences with arrangement difficulties due to the lack of availability and accessibility of interpreter services leading to long wait times.[13,44,46,50] While providing language services is a federal mandate for hospitals receiving federal funding,[2,51,52] hospital or unit culture can influence interpreter service utilization as nurses may be discouraged by leadership from using these services due to high costs with inconsistent reimbursement policies across payors. The interpreter use policies that are written may also be implemented with variable efforts and consequently have a variable impact on patient outcomes. Further, nurses have reported concerns over the lack of resources and support from their unit managers, specifically due to the lack of translators.[53] Regardless of the existence of language-translation interventions, their poor accessibility demonstrates an operational failure of the hospital. This operational failure would be alleviated through the improvement of a hospital's nursing resources. When nurses are adequately staffed, work in positive environments, and feel supported

by the institutions where they work, they may be better positioned to access and use interpreter services, if available, to overcome language barriers.

Nurse characteristics. The recruitment of underrepresented, linguistically diverse populations into the nursing workforce is one effective structural recommendation to improve care and outcomes for patients with LEP.[54] Low representation of culturally and linguistically diverse (CALD) populations within the health care workforce has been described as contributory to the lack of resources for CALD communities.[55] Through diverse hiring and recruitment, hospitals may be better equipped to provide linguistically concordant nursing care.[42,55] Hospitals' recruiting, supporting, and retaining of providers who speak the same language as their patients may contribute to better care delivery and hospital outcomes for LEP patients. However, it is important to recognize the gold standard intervention for overcoming language discordance is the use of in-person medical interpreters since they have extensive training and experience that prepares them to translate between patients and providers effectively and accurately.[56]

Nurse-managerial relationships and organizational culture. Nurses have identified managerial and organizational support as critical for the provision of culturally competent and language-concordant care; however, some nurses report feeling unsupported by the institutions where they work to adequately care for individuals with LEP. For example, one qualitative study of bilingual nurses showed that even when nurses felt comfortable providing language-concordant care to LEP patients, they were questioned or discouraged by managers for speaking to patients in a concordant language.[16] Unclear organizational language and interpretation policies, a lack of support, and general disconnect between nurses and their managers were common themes among these bilingual nurses.[16] Nurses also desired more recognition by management of the difficulties they face when caring for LEP patients.[20,57] Indifference and a lack of support by nurses' managers contributes to nurses' inability to effectively communicate with their patients by reducing their desire to do so.[20,57] Nurses have also reported a desire for improved cultural competence training by their employing institutions, such as opportunities for continuing education related to topics of culture and language.[45,57] Research has demonstrated relationships between supportive clinical practice environments, which are described as the social and organizational structure operating in the workplace, and a variety of outcomes such as higher job satisfaction for nurses, patients' decreased risk of death, and increased delivery of patient-centered care.[58–61] Examples of supportive clinical practice environments include those in which the clinical setting is favorable, clinicians have supportive working relationships, and there is substantial managerial support. Although literature has highlighted some aspects of organization culture in the context of care delivery for patients with LEP, these relationships have yet to be empirically studied.

The Influence of System-level Nursing Factors on Complex Patients

The program of studying system-level nursing factors and LEP patient outcomes is supported by well-documented associations between system-level nursing resources and

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outcomes of other complex patient populations, such as medically high-risk patients[62–64] and patients belonging to racial minority groups.[65,66] Several studies have demonstrated associations between nurse staffing levels and a range of outcomes for ICU patients, including falls, medication errors, infections, and mortality.[67–69] Better nurse work environments in ICUs are correlated with lower rates of health care-associated infections and increased patient satisfaction.[70,71] Favorable nurse work environments include factors such as supportive management, nursing leadership, and nurse-physician collaboration that respects and acknowledges the role of nurses within a health care team.[72]

Intensive care unit patients often have multifaceted medical needs that may include intubation, multi-medication regimens, severe or complicated diagnoses, and a need for close monitoring that increases the time nurses spend with them.[43,72,73] Limited English proficient patients have complex communication needs that also require additional nursing time. Thus, it is plausible that the relationships observed between hospital nursing resources and the outcomes of patients cared for in the ICU may be similar for hospitalized patients with LEP. Several studies included in our review mentioned poor nurse staffing and managerial support as key factors that interfered with the safe and effective delivery of nursing care to patients with LEP;[35,47,43] however, research on these and other hospital nursing resources in the context of patients with LEP has been extremely limited and more is needed.

Recommendations and Setting a New Research Agenda

Throughout the literature, language barriers between patients and nurses are found to add additional complexity to nursing care delivery. Without requisite system-level factors that support nurses to provide high quality patient care, nurses are hindered in their ability to establish a strong nurse-patient relationship and perform proper assessments in a timely manner—which may be an important underlying cause of the known hospital outcomes disparities for individuals with LEP. Existing health services research has shown associations between hospital nursing resources and outcomes of highly complex patients. The relationships may be similar for LEP patients. This is an important area for future empirical exploration.

In applying the QHOM framework to an examination of nursing and LEP patient outcomes, the characteristics of the hospitals and systems in which these nurses practice emerged as an important, but understudied, factor that may affect nurses' ability to provide high-quality care to LEP patients. Nurses have identified facilitators to the care delivery process for patients with LEP, such as using professional interpreters[44] or bilingual nursing staff,[16] but system-level factors such as work environments and staffing levels may be critical foundational components to facilitating or inhibiting effective use of these services.

Current research often places the responsibility for addressing language barriers on the individuals with LEP or the clinicians who care for them.[17] By applying the QHOM framework to language-barriers research, we can restructure how we think about what influences nursing care delivery for patients with LEP. Although nurses are delivering direct patient care, *system-level* factors likely moderate the relationship between care delivery and subsequent patient outcomes. Thus, interventions targeting inadequacies

at the systems level, as opposed to the clinician level, may facilitate clinicians' abilities to deliver high-quality LEP patient care. Such systems-level interventions may include ensuring adequate nursing staffing and supportive work environments. Therefore, these system-level factors are worth examining in association with outcomes of patients with LEP, and such future research may be more likely than prior recommendations to bring about change toward achieving equitable health outcomes.

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References

1. U.S. Census Bureau. American Community Survey Data. Washington, DC: U.S. Census Bureau, 2023. Available at https://www.census.gov/programs-surveys/acs/data.html.
2. Civil Rights Division. Title VI Of The Civil Rights Act Of 1964 42 U.S.C. § 2000d Et Seq. Washington DC: U.S. Department of Justice, 2023. Available at https://www.justice.gov/crt/fcs/TitleVI-Overview.
3. Ashing KT, George M, Jones V. Health-related quality of life and care satisfaction outcomes: Informing psychosocial oncology care among Latina and African-American young breast cancer survivors. Psychooncology. 2018 Apr;27(4):1213–20. Epub 2018 Feb 13.
https://doi.org/10.1002/pon.4650
PMid:29370454
4. Chung S, Huang Q, LaMori J, et al. Patient-reported experiences in discussing prescribed medications with a health care provider: evidence for racial/ethnic disparities in a large health care delivery system. Popul Health Manag. 2020 Feb;23(1):78–84. Epub 2019 Apr 23.
https://doi.org/10.1089/pop.2018.0206
PMid:31013464
5. Tang EW, Go J, Kwok A, et al. The relationship between language proficiency and surgical length of stay following cardiac bypass surgery. Eur J Cardiovasc Nurs. 2016 Oct;15(6):438–46. Epub 2015 Jul 21.
https://doi.org/10.1177/1474515115596645
PMid:26198643
6. John-Baptiste A, Naglie G, Tomlinson G, et al. The effect of English language proficiency on length of stay and in-hospital mortality. J Gen Intern Med. 2004 Mar;19(3):221–8.
https://doi.org/10.1111/j.1525-1497.2004.21205.x
PMid:15009776
7. Squires A, Ma C, Miner S, et a;. Assessing the influence of patient language preference on 30 day hospital readmission risk from home health care: a retrospective analysis. Int J Nurs Stud. 2022 Jan:125:104093. Epub 2021 Oct 1.
https://doi.org/10.1016/j.ijnurstu.2021.104093
PMid:34710627
8. Karliner LS, Kim SE, Meltzer DO, et al. Influence of language barriers on outcomes of hospital care for general medicine inpatients. J Hosp Med. 2010 May–Jun;5(5):276–82.

[130.91.156.128] Project MUSE (2024-02-27 13:52 GMT) University Of Pennsylvania Libraries

https://doi.org/10.1002/jhm.658
PMid:20533573
9. Batista R, Prud'homme D, Rhodes E, et al. Quality and safety in long-term care in Ontario: the impact of language discordance. J Am Med Dir Assoc. 2021 Oct;22(10): 2147–2153.e3. Epub 2021 Jan 9.
https://doi.org/10.1016/j.jamda.2020.12.007
PMid:33434567
10. Miner SM, Squires AP, Ma C, et al. detecting disparities in medication management among limited English proficient and English proficient home health patients. Home Health Care Manag Pract. 2020 Feb;32(1):28–33.
https://doi.org/10.1177/1084822319865546
11. Rogers AJ, Delgado CA, Simon HK. The effect of limited English proficiency on admission rates from a pediatric ED: stratification by triage acuity. Am J Emerg Med. 2004 Nov;22(7):534–6.
https://doi.org/10.1016/j.ajem.2004.08.012
PMid:15666256
12. Schwartz HEM, Matthay ZA, Menza R, et al. Inequity in discharge pain management for trauma patients with limited English proficiency. J Trauma Acute Care Surg. 2021 Nov 1;91(5):898–902.
https://doi.org/10.1097/TA.0000000000003294
PMid:34039923
13. Jones SM. Emergency nurses' caring experiences with Mexican American patients. J Emerg Nurs. 2008 Jun;34(3):199–204.
https://doi.org/10.1016/j.jen.2007.05.009
PMid:18558248
14. Coleman J-S, Angosta AD. The lived experiences of acute-care bedside registered nurses caring for patients and their families with limited English proficiency: a silent shift. J Clin Nurs. 2017 Mar;26(5–6):678–89. Epub 2016 Nov 28.
https://doi.org/10.1111/jocn.13567
PMid:27571582
15. Karliner LS, Jacobs EA, Chen AH, et al. Do professional interpreters improve clinical care for patients with limited English proficiency? A systematic review of the literature. Health Serv Res. 2007 Apr;42(2):727–54.
https://doi.org/10.1111/j.1475-6773.2006.00629.x
PMid:17362215
16. Ali PA, Johnson S. Speaking my patient's language: bilingual nurses' perspective about provision of language concordant care to patients with limited English proficiency. J Adv Nurs. 2017 Feb;73(2):421-32. Epub 2016 Sep 29.
https://doi.org/10.1111/j.1466-7657.2008.00697.x
PMid:19646175
17. Betancourt JR, Renfrew MR, Green AR, et al. Improving patient safety systems for patients with limited English proficiency: a guide for hospitals. Rockville, MD: Agency for Healthcare Research and Quality, 2012. Available at https://www.ahrq.gov/sites /default/files/publications/files/lepguide.pdf.
18. Anoosheh M, Zarkhah S, Faghihzadeh S, et al. Nurse-patient communication barriers in Iranian nursing. Int Nurs Rev. 2009 Jun;56(2):243–9.
https://doi.org/10.1111/j.1466-7657.2008.00697.x
PMid:19646175
19. Valizadeh L, Zamanzadeh V, Ghahramanian A, et al. Factors influencing nurse-

to-parent communication in culturally sensitive pediatric care: a qualitative study. Contemp Nurse. 2017 Aug;53(4):474–88. Epub 2017 Dec 5.
https://doi.org/10.1080/10376178.2017.1409644
PMid:29169308

20. Chae D, Park Y. Organisational cultural competence needed to care for foreign patients: a focus on nursing management. J Nurs Manag. 2019 Jan;27(1):197–206. Epub 2018 Sep 10.
https://doi.org/10.1111/jonm.12665
PMid:30198599
21. Aiken LH, Cerón C, Simonetti M, et al. Hospital nurse staffing and patient outcomes. Revista Médica Clínica Las Condes. 2018 May–Jun;29(3):322–7.
https://doi.org/10.1016/j.rmclc.2018.04.011
22. Lasater KB, Mchugh MD. Nurse staffing and the work environment linked to readmissions among older adults following elective total hip and knee replacement. Int J Qual Health Care. 2016 Apr;28(2):253–8. Epub 2016 Feb 2.
https://doi.org/10.1093/intqhc/mzw007
PMid:26843548
23. Patel AT, Lee BR, Donegan R, et al. Length of stay for patients with limited English proficiency in pediatric urgent care. Clin Pediatr (Phila). 2020 May;59(4–5):421–8. Epub 2020 Jan 29.
https://doi.org/10.1177/0009922820902439
PMid:31994413
24. Mitchell PH, Ferketich S, Jennings BM. Quality Health Outcomes Model. American Academy of Nursing Expert Panel on Quality Health Care Image J Nurs Sch. 1998;30(1):43–6.
https://doi.org/10.1111/j.1547-5069.1998.tb01234.x
PMid:9549940
25. de Moissac D, Bowen S. Impact of language barriers on quality of care and patient safety for official language minority Francophones in Canada. J Patient Exp. 2019 Mar;6(1):24–32. Epub 2018 Apr 18.
https://doi.org/10.1177/2374373518769008
PMid:31236448
26. Pun JKH, Chan EA, Murray KA, et al. Complexities of emergency communication: clinicians' perceptions of communication challenges in a trilingual emergency department. J Clin Nurs. 2017 Nov;26(21–22):3396–407. Epub 2017 Mar 3.
https://doi.org/10.1111/jocn.13699
PMid:28001321
27. van Rosse F, de Bruijne M, Suurmond J, et al. Language barriers and patient safety risks in hospital care. a mixed methods study. Int J Nurs Stud. 2016 Feb:54:45–53. Epub 2015 Mar 25.
https://doi.org/10.1016/j.ijnurstu.2015.03.012
PMid:25840899
28. Divi C, Koss RG, Schmaltz SP, et al. Language proficiency and adverse events in US hospitals: a pilot study. Int J Qual Health Care. 2007 Apr;19(2):60–7. Epub 2007 Feb 2.
https://doi.org/10.1093/intqhc/mzl069
PMid:17277013
29. Rodriguez F, Hong C, Chang Y, et al. Limited English proficient patients and time spent in therapeutic range in a warfarin anticoagulation clinic. J Am Heart Assoc. 2013 Jul 5;2(4):e000170.

[130.91.156.128] Project MUSE (2024-02-27 13:52 GMT) University Of Pennsylvania Libraries

https://doi.org/10.1161/JAHA.113.000170
PMid:23832325

30. Flores G, Abreu M, Barone CP, et al. Errors of medical interpretation and their potential clinical consequences: a comparison of professional versus ad hoc versus no interpreters. Ann Emerg Med. 2012 Nov;60(5):545–53. Epub 2012 Mar 15.
https://doi.org/10.1016/j.annemergmed.2012.01.025
PMid:22424655

31. Lindholm M, Hargraves JL, Ferguson WJ, et al. Professional language interpretation and inpatient length of stay and readmission rates. J Gen Intern Med. 2012 Oct;27(10):1294–9. Epub 2012 Apr 18.
https://doi.org/10.1007/s11606-012-2041-5
PMid:22528618

32. Gardam M, Verma G, Campbell A, et al. Impact of the patient-provider relationship on the survival of foreign born outpatients with tuberculosis. J Immigr Minor Health. 2009 Dec;11(6):437–45. Epub 2009 Jan 6.
https://doi.org/10.1007/s10903-008-9221-8
PMid:19127431

33. Sobel LL, Metzler Sawin E. Guiding the process of culturally competent care with Hispanic patients. J Transcult Nurs. 2016 May;27(3):226–32. Epub 2014 Nov 19.
https://doi.org/10.1177/1043659614558452
PMid:25409673

34. Kirca N, Bademli K. Relationship between communication skills and care behaviors of nurses. Perspect Psychiatr Care. 2019 Oct;55(4):624–31. Epub 2019 Apr 16.
https://doi.org/10.1111/ppc.12381
PMid:30990906

35. Norouzinia R, Aghabarari M, Shiri M, et al. Communication barriers perceived by nurses and patients. Glob J Health Sci. 2015 Sep 28;8(6):65–74.
https://doi.org/10.5539/gjhs.v8n6p65
PMid:26755475

36. Degni F, Suominen S, Essén B, et al. Communication and cultural issues in providing reproductive health care to immigrant women: health care providers' experiences in meeting the needs of [corrected] Somali women living in Finland. J Immigr Minor Health. 2012 Apr;14(2):330–43.
https://doi.org/10.1007/s10903-011-9465-6
PMid:21465142

37. Stephen JM. Pediatric nurses' experiences in caring for non–English speaking patients and families. J Transcult Nurs. 2021 Nov;32(6):690–6. Epub 2021 Jan 15.
https://doi.org/10.1177/1043659620986607
PMid:33446063

38. Kalisch BJ, Xie B. Errors of omission: missed nursing care. West J Nurs Res. 2014 Aug;36(7):875–90. Epub 2014 Apr 29.
https://doi.org/10.1177/0193945914531859
PMid:24782432

39. Recio-Saucedo A, Dall'Ora C, Maruotti A, et al. What impact does nursing care left undone have on patient outcomes? Review of the literature. J Clin Nurs. 2018 Jun;27(11–12):2248–59. Epub 2017 Oct 16.
https://doi.org/10.1111/jocn.14058
PMid:28859254

40. He M, Zhu X, Dong Y, et al. Exploring the role of communication in missed nursing care: A systematic review. J Adv Nurs. 2022 Dec;78(12):4019–33. Epub 2022 Sep 12.
https://doi.org/10.1111/jan.15444
PMid:36097637
41. Reppas-Rindlisbacher C, Panov ED, Cuperfain AB, et al. A survey of nurses' perspectives on delirium screening in older adult medical inpatients with limited English proficiency. J Gerontol Nurs. 2021 Apr;47(4):29–34. Epub 2021 Apr 1.
https://doi.org/10.3928/00989134-20210309-05
PMid:34038248
42. Gelete BA, Radie YT, Areri HA. Exploring cultural competence of nurses and associated barriers in Ethiopian tertiary hospitals: A qualitative study. Appl Nurs Res. 2021 Dec:62:151511. Epub 2021 Oct 6.
https://doi.org/10.1016/j.apnr.2021.151511
PMid:34815007
43. Farley R, Askew D, Kay M. Caring for refugees in general practice: perspectives from the coalface. Aust J Prim Health. 2014;20(1):85–91.
https://doi.org/10.1071/PY12068
PMid:23257393
44. Tam I, Huang MZ, Patel A, et al. Spanish interpreter services for the hospitalized pediatric patient: provider and interpreter perceptions. Acad Pediatr. 2020 Mar;20(2): 216–24. Epub 2019 Aug 22.
https://doi.org/10.1016/j.acap.2019.08.012
PMid:31445969
45. Starr S, Wallace DC. Self-reported cultural competence of public health nurses in a southeastern U.S. public health department. Public Health Nurs. 2009 Jan–Feb;26(1):48–57.
https://doi.org/10.1111/j.1525-1446.2008.00753.x
PMid:19154192
46. Nkulu Kalengayi FK, Hurtig AK, Ahlm C, et al. "It is a challenge to do it the right way": an interpretive description of caregivers' experiences in caring for migrant patients in Northern Sweden. BMC Health Serv Res. 2012 Nov 29:12:433.
https://doi.org/10.1186/1472-6963-12-433
PMid:23194441
47. Ali PA, Watson R. Language barriers and their impact on provision of care to patients with limited English proficiency: nurses' perspectives. J Clin Nurs. 2018 Mar;27(5–6):e1152–60.
https://doi.org/10.1111/jocn.14204
48. Norton-Westwood D, Robertson-Malt S, Anderson R. A randomized controlled trial to assess the impact of an Admission Service on patient and staff satisfaction. Int J Nurs Pract. 2010 Oct;16(5):461–71.
https://doi.org/10.1111/j.1440-172X.2010.01870.x
PMid:20854343
49. Marcus J, Clark PC, MacMillan D, Ketchie M. Nurses' perceptions of technology used in language interpretation for patients with limited English proficiency: nurses prefer video remote interpretation to telephone-based technology. Am J Nurs. 2020 Sep;120(9):48–58.
https://doi.org/10.1097/01.NAJ.0000697652.73975.37
PMid:32858699

[130.91.156.128] Project MUSE (2024-02-27 13:52 GMT) University Of Pennsylvania Libraries

50. Diamond L, Izquierdo K, Canfield D, et al. A systematic review of the impact of patient–physician non-English language concordance on quality of care and outcomes. J Gen Intern Med. 2019 Aug;34(8):1591–606. Epub 2019 May 30.
https://doi.org/10.1007/s11606-019-04847-5
PMid:31147980
51. Rajbhandari P, Garber MD. Improving language equity in hospitalized children of families with lep: challenges and solutions. Hosp Pediatr. 2022 Oct 1;12(10):e367–73.
https://doi.org/10.1542/hpeds.2022-006537
PMid:36124635
52. Al Shamsi H, Almutairi AG, Al Mashrafi S, et al. Implications of language barriers for healthcare: a systematic review. Oman Med J. 2020 Apr 30;35(2):e122.
https://doi.org/10.5001/omj.2020.40
PMid:32411417
53. Giltenane M, Sheridan A, Kroll T, et al. Work environment challenging Irish public health nurses' care quality: First postnatal visit. Public Health Nurs. 2022 Jan;39(1): 202–13. Epub 2021 Oct 20.
https://doi.org/10.1111/phn.12993
PMid:34672017
54. Carter-Pokras O, Brown P, Martinez I, et al. Latin American—trained nurse perspective on Latino health disparities. J Transcult Nurs. 2008 Apr;19(2):161–6. Epub 2008 Feb 8.
https://doi.org/10.1177/1043659607312972
PMid:18263845
55. Komaric N, Bedford S, van Driel ML. Two sides of the coin: patient and provider perceptions of health care delivery to patients from culturally and linguistically diverse backgrounds. BMC Health Serv Res. 2012 Sep 18:12:322.
https://doi.org/10.1186/1472-6963-12-322
PMid:22985266
56. Rimmer A. Can patients use family members as non-professional interpreters in consultations? BMJ. 2020 Feb 11:368:m447.
https://doi.org/10.1136/bmj.m447
PMid:32047051
57. Farahani MA, Sahragard R, Carroll JK, et al. Communication barriers to patient education in cardiac inpatient care: a qualitative study of multiple perspectives. Int J Nurs Pract. 2011 Jun;17(3):322–8.
https://doi.org/10.1111/j.1440-172X.2011.01940.x
PMid:21605274
58. Poghosyan L, Liu J, Norful AA. Nurse practitioners as primary care providers with their own patient panels and organizational structures: a cross-sectional study. Int J Nurs Stud. 2017 Sep:74:1–7. Epub 2017 May 25.
https://doi.org/10.1016/j.ijnurstu.2017.05.004
PMid:28577459
59. Poghosyan L, Nannini A, Clarke S. Organizational climate in primary care settings: implications for nurse practitioner practice. J Am Assoc Nurse Pract. 2013 Mar;25(3): 134–40. Epub 2012 Aug 16.
https://doi.org/10.1111/j.1745-7599.2012.00765.x
PMid:24218200
60. Margo Brooks Carthon J, Brom H, Poghosyan L, et al. Supportive clinical practice envi-

ronments associated with patient-centered care. J Nurse Pract. 2020 Apr;16(4):294–8. Epub 2020 Mar 11.
https://doi.org/10.1016/j.nurpra.2020.01.019
PMid:32863798

61. Aiken LH, Clarke SP, Sloane DM, et al. Effects of hospital care environment on patient mortality and nurse outcomes. J Nurs Adm. 2009 Jul–Aug;39(7–8 Suppl):S45–51.
https://doi.org/10.1097/NNA.0b013e3181aeb4cf
PMid:19641438

62. Lasater KB, McHugh MD, Rosenbaum PR, et al. Evaluating the costs and outcomes of hospital nursing resources: a matched cohort study of patients with common medical conditions. J Gen Intern Med. 2021 Jan;36(1):84–91. Epub 2020 Aug 31.
https://doi.org/10.1007/s11606-020-06151-z
PMid:32869196

63. Lasater KB, McHugh M, Rosenbaum PR, et al. Valuing hospital investments in nursing: multistate matched-cohort study of surgical patients. BMJ Qual Saf. 2021 Jan;30(1): 46–55. Epub 2020 Mar 27.
https://doi.org/10.1136/bmjqs-2019-010534
PMid:32220938

64. Silber JH, Rosenbaum PR, McHugh MD, et al. Comparison of the value of nursing work environments in hospitals across different levels of patient risk. JAMA Surg. 2016 Jun 1;151(6):527–36.
https://doi.org/10.1001/jamasurg.2015.4908
PMid:26791112

65. Brooks Carthon JM, Kutney-Lee A, Jarrín O, et al. Nurse staffing and postsurgical outcomes in Black adults. J Am Geriatr Soc. 2012 Jun;60(6):1078–84.
https://doi.org/10.1111/j.1532-5415.2012.03990.x
PMid:22690984

66. Brooks Carthon M, Brom H, McHugh M, et al. Better nurse staffing is associated with survival for Black patients and diminishes racial disparities in survival after in-hospital cardiac arrests. Med Care. 2021 Feb 1;59(2):169–76.
https://doi.org/10.1097/MLR.0000000000001464
PMid:33201082

67. Aragon Penoyer D. Nurse staffing and patient outcomes in critical care: A concise review. Crit Care Med. 2010 Jul;38(7):1521–8
https://doi.org/10.1097/CCM.0b013e3181e47888
PMid:20473146

68. Whitman GR, Kim Y, Davidson LJ, et al. The impact of staffing on patient outcomes across specialty units. J Nurs Adm. 2002 Dec;32(12):633–9.
https://doi.org/10.1097/00005110-200212000-00008
PMid:12483084

69. West E, Barron DN, Harrison D, et al. Nurse staffing, medical staffing and mortality in Intensive Care: an observational study. Int J Nurs Stud. 2014 May;51(5):781–94. Epub 2014 Feb 27.
https://doi.org/10.1016/j.ijnurstu.2014.02.007
PMid:24636667

70. Boev C. The relationship between nurses' perception of work environment and patient satisfaction in adult critical care. J Nurs Scholarsh. 2012 Dec;44(4):368–75. Epub 2012 Sep 18.

https://doi.org/10.1111/j.1547-5069.2012.01466.x
PMid:22989120

71. Krupp A, Lasater KB, McHugh MD. Intensive care unit utilization following major surgery and the nurse work environment. AACN Adv Crit Care. 2021 Dec 15;32(4):381–90.
https://doi.org/10.4037/aacnacc2021383
PMid:34879139

72. Lake ET, Sanders J, Duan R, et al. A meta-analysis of the associations between the nurse work environment in hospitals and 4 sets of outcomes. Med Care. 2019 May;57(5):353–61.
https://doi.org/10.1097/MLR.0000000000001109
PMid:30908381

73. Magnus VS, Turkington L. Communication interaction in ICU—Patient and staff experiences and perceptions. Intensive Crit Care Nurs. 2006 Jun;22(3):167–80. Epub 2005 Nov 17.
https://doi.org/10.1016/j.iccn.2005.09.009
PMid:16298132